



2.1.6 Wrap-Up: Cool Down

Determine an equivalent expression for each algebraic expression using the laws of exponents. Show your work.

1. $(56g^7)\left(\frac{4}{5}g^{-2}\right)$

2. $(-x^2y)(7.3x^8y^{11})(0.1x^9y^3)^2$



Unit 2, Lesson 2: Applying Quotient Laws of Exponents to Algebraic Expressions



Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.



Learning Target

- ☐ I can generate equivalent algebraic expressions to find the quotient of two monomials using the properties (laws) of exponents.



2.2.1 Warm-Up: Would You Rather?

1. Would you rather earn a salary of \$50,000 per year, while everyone else earns \$25,000, or earn a salary of \$100,000 per year, while everyone else earns \$200,000?



2. Explain to your partner how you decided on the option you chose.



2.2.2 Guided Instruction: Quotient of Powers

When dividing monomials, the **quotient of powers** law of exponents is used.

A breakdown of dividing 3^5 by 3^4 is shown.

$$\frac{3^5}{3^4} = \frac{\cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot 3}{\cancel{3} \cdot \cancel{3} \cdot \cancel{3} \cdot \cancel{3}} = \frac{3}{1} = 3^1 = 3$$

A breakdown of dividing x^5 by x^2 is shown.

$$\frac{x^5}{x^2} = \frac{\cancel{x} \cdot \cancel{x} \cdot x \cdot x \cdot x}{\cancel{x} \cdot \cancel{x}} = \frac{x \cdot x \cdot x}{1} = x^3$$

1. Consider the expression $\frac{x^2}{x^5}$.
 - a. Complete the steps to find an equivalent expression.

$$\frac{x^2}{x^5} = \frac{x \cdot x}{x \cdot x \cdot x \cdot x \cdot x} = \underline{\hspace{2cm}}$$

- b. With your partner, discuss what value is in the numerator of the equivalent expression.



Quotient of Powers

$$\frac{a^m}{a^n} = a^{m-n} \text{ and conversely } a^{m-n} = \frac{a^m}{a^n}$$

- c. Use the quotient of powers law to find an equivalent expression for $\frac{x^2}{x^5}$. Show your work.

- d. Discuss with your partner how the expression found using the quotient of powers and the expression found when rewriting $\frac{x^2}{x^5}$ as $\frac{x \cdot x}{x \cdot x \cdot x \cdot x \cdot x}$ are the same.

2. Consider the expression $\frac{x^3 y^{11} z^3}{x^2 y^2}$.

- a. Use the property of exponents to find an equivalent expression.

$$\frac{x^3 y^{11} z^3}{x^2 y^2} = x^{(3-2)} y^{(11-2)} z^{(3)} = x \boxed{} y \boxed{} z^3$$

- b. Discuss with your partner why the power of z did not change.

3. Consider the expression $\frac{12x^3 y^1}{6x^1 y^3}$.

- a. Determine an expression equivalent to $\frac{12x^3 y^1}{6x^1 y^3}$.

$$\left(\frac{12}{6}\right) \left(\frac{x^3}{x^1}\right) \left(\frac{y^1}{y^3}\right) =$$

- b. The variable y is in the denominator of the expression equivalent to $\frac{12x^3 y^1}{6x^1 y^3}$. Discuss with your partner what value(s), when substituted in for y , would make the expression undefined.

Domain restrictions should be written to exclude the value(s) which would make the expression undefined. For example, in the expression $\frac{1}{x^3}$, when 0 is substituted for x the expression is undefined, making 0 a domain restriction. The restriction for $\frac{1}{x^3}$ is represented as $x \neq 0$.

- c. Write the domain restriction for the expression, $\frac{12x^3y^1}{6x^1y^3}$.



2.2.3 Guided Practice: Generating Equivalent Algebraic Expressions Using Quotient of Powers

1. Complete the table for each expression.

Expression	Equivalent Expression	Domain Restriction?	If yes, state the restriction.
$\frac{12s^5t^9}{20t^3}$		☐ yes ☐ no	
$\frac{21a^8b^3c^{15}}{7a^3b^{12}c^2}$		☐ yes ☐ no	
$\frac{3nm^4}{6n^2m^7}$		☐ yes ☐ no	



2.2.4 Guided Instruction: Power of a Quotient

1. Three students generated equivalent expressions using exponent laws.
 - a. Finish Amir's work using a similar strategy to the one used by Luis and Maya.

Luis	Maya	Amir
$\left(\frac{3}{11}\right)^5$	$\left(\frac{x}{6y}\right)^2$	$\left(\frac{7c}{r}\right)^3$
$\frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11} \cdot \frac{3}{11}$	$\frac{x}{6y} \cdot \frac{x}{6y}$	$\frac{7c}{r} \cdot$
$\frac{3 \cdot 3 \cdot 3 \cdot 3 \cdot 3}{11 \cdot 11 \cdot 11 \cdot 11 \cdot 11}$	$\frac{x \cdot x}{6 \cdot 6 \cdot y \cdot y}$	
$\frac{3^5}{11^5}$	$\frac{x^2}{36y^2}$	

- b. Complete the table.

Expression	Equivalent Expression	Domain Restriction?	If yes, state the restriction.
$\left(\frac{3}{11}\right)^5$	$\frac{3^5}{11^5}$	☐ yes ☐ no	
$\left(\frac{x}{6y}\right)^2$	$\frac{x^2}{36y^2}$	☐ yes ☐ no	
$\left(\frac{7c}{r}\right)^3$		☐ yes ☐ no	

2. Complete the statement.

When a and b are nonzero real numbers, and m is an integer, the exponential expression $\left(\frac{a}{b}\right)^m$ is equal to

- ☐ a^{b-m} .
☐ $\frac{a^m}{b^m}$.
☐ $a^m - b^m$.



Power of a Quotient

$\left(\frac{a}{b}\right)^m = \frac{a^m}{b^m}$ and conversely $\frac{a^m}{b^m} = \left(\frac{a}{b}\right)^m$



2.2.5 Activity: Matching Equivalent Expressions

- Match each expression to an equivalent expression by drawing an arrow connecting them.

Expression	Equivalent Expression
$\frac{1a^4y^4}{256b^4p^4}$	$\frac{a^2}{-4}$
$\left(\frac{8ay}{3bp}\right)^3$	$-2a^3b$
$\frac{-18a^{12}by^4}{9a^9y^4}$	$\frac{100a^6p^7y}{55ab^3p^{13}y}$
$\frac{1}{a^5b^4}$	$\left(\frac{ay}{4bp}\right)^4$
$\frac{20a^5}{11b^3p^6}$	$\frac{a^{20}b^7p^0}{a^{25}b^{11}}$
$\frac{a^2b^3}{-4b^3}$	$\frac{512a^3y^3}{27b^3p^3}$



2.2.6 Your Turn!

- Find the quotient of each expression by using the laws of exponents to generate an equivalent expression.

Expression	Equivalent Expression
a. $\frac{s^2t^{18}}{-5st^7}$	
b. $\left(\frac{0.5d}{10mg}\right)^2$	
c. $\frac{-204u^2w^8r^{31}}{17u^{16}w^{15}r^4}$	
d. $\left(\frac{e}{k^5}\right)^4$	
e. $\frac{49p^4h^8}{81p^{10}}$	



2.2.7 Wrap-Up: Reflection

1. Choose the emoji that reflects your understanding of generating equivalent algebraic expressions to find quotient of two monomials using the properties (laws) of exponents.



0 1 2 3 4 5 6 7 8 9 10

2. Complete the statement.

I chose this emoji because _____



Unit 2, Lesson 3: Applying Exponent Laws to Generate Equivalent Algebraic Expressions



Benchmark

MA.912.NSO.1.2 Generate equivalent algebraic expressions using the properties of exponents.



Learning Target

- ☐ I can generate equivalent algebraic expressions using properties (laws) of exponents.